

Subject Description Form

Subject Code	ABCT1701
Subject Title	Fundamental Chemical Science
Credit Value	3
Level	1
Prerequisite	No prerequisite.
Exclusion	ABCT1700 and ABCT1701 are mutually exclusive.
Objectives	This subject introduces the foundational principles of chemical science, with a molecular perspective for understanding the structure, properties, and transformations of matter. Students will develop quantitative problem-solving skills and chemical literacy.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand atomic structure, periodic trends, chemical bonding, molecular geometry, intermolecular forces, and basic thermochemical concepts as foundations of chemical science. - Apply appropriate chemical symbols, formulas, equations, and nomenclature to represent elements, compounds, and chemical reactions accurately. - Appreciate the relevance of fundamental chemical science to further studies in science and technology and to issues in modern life, such as materials, environment, and health
Subject Synopsis/ Indicative Syllabus	1. Atomic structure and the periodic table - Atoms and the periodic table - Periodic classification and properties of elements. 2. Chemical formulas, nomenclature and stoichiometry - Chemical formulas and nomenclature - Stoichiometry and chemical calculations 3. Chemical reactions and energy - Types of chemical reactions, classification and representation - Thermochemistry and introductory concepts of energy conservation in reactions. 4. Gases, electronic structure and periodic trends - Gases and kinetic molecular theory - Electronic structure and periodic trends

	5. Bonding, molecular structure and condensed phases - Chemical bonding and molecular geometry - Intermolecular forces and condensed phases - Solutions and concentration 6. Equilibria and acid–base chemistry - Chemical equilibrium in chemical systems. - Acids, bases, and pH and simple equilibrium calculations for aqueous acid–base systems.				
Teaching and Learning Methodology	Lecture: The fundamental principles of chemistry will be explained. Examples will be used to illustrate the concepts and ideas in the lecture. Take-home problem sets will be given, and the students are encouraged to solve the problems before seeking assistance. Tutorial: Students present their solutions on a set of problems in the tutorial. Students should try the problems before seeking assistance. These problem sets provide them opportunities to apply the knowledge gained from the lecture. They also help the students consolidate and familiarize with what they have learned. Furthermore, students can develop a deeper understanding of the subject through group discussion and self-study.				
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods / tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)		
			a	b	c
	Test 1	20%	✓	✓	
	Test 2	20%	✓	✓	
	Assignments	5%	✓	✓	✓
	In-class pop-up quizzes	5%	✓		✓
	Examination	50%	✓	✓	✓
	Total	100%			
	<i>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:</i> In-class pop-up quizzes, tests and examination are used to assess student's learning in key physical concepts in atomic structure, chemical bonding, and chemical reactions. Homework assignments (e.g., end-of-chapter exercises and online assignments) would reinforce student's knowledge in these key topics and practice for their numerical skills and problem-solving skill through analysis of experimental data.				

Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Tutorial	6 Hrs.
	Other student study effort:	
	▪ Self-study	71 Hrs.
	▪ Problem assignments/homework	16 Hrs.
	Total student study effort	126 Hrs.
Reading List and References	<u>Essential Reading</u> ▪ Tro, N. J., Introductory Chemistry Essentials, Pearson, 2018.	